AKSHAY SURESH

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Physicist and ML researcher building AI and signal processing systems for scientific discovery. Published 9 refereed astronomy papers (7 first-author) on signal detection in terabyte-scale radio data. Shipped production ML systems, including geospatial foundation models, at AI startups in London.

EDUCATION

MS & PhD, Cornell University, USA — 08/2023
Astronomy major, Physics minor — GPA: 4.0/4.0

BS & MS Dual Degree with **Institute Gold Medal**, IISER Pune, India — 05/2017
Physics major, Mathematics minor — GPA: 9.9/10

AI FOR SCIENCE RESEARCH EXPERIENCE

Graduate Researcher, Cornell University, USA — 08/2017 – 08/2023

- **Developed BLIPSS, an open-source package** to search for periodic radar-like transmissions. A pilot Galactic Center survey limited such transmitters to fewer than one in 600,000 stars. [AJ 2023; covered by Reuters]
- **Led a three-person proof-of-principle project training CNNs of varying depth in PyTorch** to classify radio-frequency interference by spectrotemporal morphology.
- Engineered an automated, memory-efficient pipeline for parallel processing of 10 TB of data at over 500 GB/hr on supercomputing platforms.

Machine Learning Researcher, Frontier Development Lab USA — 06/2022 – 08/2022

- **Built an LSTM model yielding 70–90% accurate short-term forecasts** of induced earthquake counts from underground carbon storage, complementing the team's physics-based model.
- Co-developed a streamlined physics-based forecasting model that cut training time from 22 hours on a supercomputer to 3 minutes on a tablet. **Team won the FDL 2022 Most Useful Contribution Award.**

INDUSTRY EXPERIENCE

Machine Learning Engineer (full-time), Treefera — 07/2025 – 05/2026

- **Deployed an open-source geospatial foundation model** for forest biomass quantification, scaling coverage from local to global in 8 weeks.
- Conceptualized and built an internal software toolkit shared across company projects, shortening initial development time by 15–25% per new initiative.

Data Science Fellow (contract), Faculty AI — 05/2025 – 07/2025

- **Implemented a retrieval-augmented framework for granular skill evaluation and scalable candidate feedback generation**, cutting recruitment costs by up to 30% per new hire.

Independent AI Research Consultant — 01/2024 – 04/2025

- **Executed test-driven development of a U-Net model** to segment Amazon rainforest cover in optical satellite imagery, reaching 97% recall and 0.94 IoU on held-out test data.
- Designed the technical roadmap for an early-stage startup building a data-as-a-service platform, empowering consumers to save up to 15% on their monthly electricity bills through tailored recommendations.

TECHNICAL SKILLS

Deep Learning	Geospatial foundation models, CNNs/U-Net, LSTMs, RAG, prompt engineering
Scientific Computing	Signal processing, time-series forecasting, high-performance computing
Programming Languages	Python (PyTorch, sklearn, numpy), SQL, bash, LaTeX
Software Engineering	Production code development, CI/CD (CircleCI, GitHub Actions), GCP, AWS